Subject: ENVIRONMENTAL IMPACT ASSESSMENT

Subject Code: 23A01504c

UNIT – V

Environmental Acts and Notifications

I. Introduction – Environmental Legislation and its Objectives

Environmental legislation means the laws made by the government to protect the environment, control pollution and conserve natural resources.

- **Constitutional basis** Article 48A (Directive Principle) – the State shall protect and improve the environment and safeguard forests and wildlife. Article 51A(g) (Fundamental Duty) – every citizen shall protect and improve the natural environment. Article 21 gives the right to a clean environment (Supreme Court judgments).
- Stockholm Conference 1972 on the Human Environment led India to make many environmental laws.

Objectives of environmental legislation

- To prevent, control and abate pollution of air, water, soil and noise.
- To conserve forests, wildlife and biodiversity.
- To fix standards for emission and effluent discharge.
- To give powers to authorities to inspect, monitor and punish violations.
- To ensure sustainable use of natural resources.
- To make industries responsible (polluter pays principle) and to promote public participation.
- To bring EIA and environmental audit under legal control.

Year	Law
1972	Wild Life (Protection) Act
1974	Water (Prevention and Control of Pollution) Act

1980	Forest (Conservation) Act
1981	Air (Prevention and Control of Pollution) Act
1986	Environment (Protection) Act
1991	Public Liability Insurance Act
2002	Biological Diversity Act
2010	National Green Tribunal Act

2. The Environment (Protection) Act, 1986

- Enacted after the Bhopal gas tragedy (1984), under Article 253 of the Constitution to implement the decisions of the Stockholm Conference. Came into force on 19 November 1986.
- It is an umbrella (framework) Act that fills the gaps in the earlier Acts and gives wide powers to the Central Government.
- Has 26 sections divided into 4 chapters.
- **Definitions (Section 2)** environment includes water, air, land and their inter-relationship with human beings, other living creatures, plants, micro-organisms and property. Environmental pollutant is any solid, liquid or gaseous substance present in such concentration as may be injurious to the environment.

Important provisions

- **Section 3** Central Government can take all measures for protecting and improving the environment, including setting up authorities, fixing standards, restricting industrial locations and laying down procedures for safeguards against accidents.
- **Section 5** power to issue directions, including closure, prohibition or regulation of any industry, and stopping electricity and water supply.
- **Section 6** power to make rules for standards of quality, maximum limits of

pollutants, handling of hazardous substances, restricted areas and so on.

- **Section 7** no person shall emit or discharge pollutants above the prescribed standards.
- **Section 8** hazardous substances must be handled as per procedures and safeguards.
- **Section 10** powers of entry and inspection; Section 11 :: power to take samples.
- **Section 15** penalty for contravention - imprisonment up to 5 years, or fine up to Rs. 1 lakh, or both; continuing violation extra fine of Rs. 5000 per day; after one year of contravention the imprisonment may extend to 7 years.
- **Section 19** court can take notice of a complaint made by the Government, or by any person after giving 60 days notice.
- **Section 17** offences by Government departments and Section 16 :: offences by companies (the person in charge is responsible).

Rules made under the Act

- **Environment (Protection) Rules 1986** standards for emission and effluent.
- EIA Notification 1994 (replaced in 2006), Hazardous Waste Rules, Biomedical Waste Rules, Noise Pollution Rules 2000, Coastal Regulation Zone Notification, Environmental Statement (1993).

3. The Water (Prevention and Control of Pollution) Act, 1974

- **Objective** prevention and control of water pollution and maintaining or restoring the wholesomeness of water. It is the first major law on pollution in India.
- **Water cess Act 1977** tax on industries using water, used to finance the Boards.
- Water pollution means contamination of water or change in its properties, or discharge of any sewage, trade effluent or other substance which is likely to

cause a nuisance or harm to public health, animals, plants or aquatic life.

Institutions

- **Central Pollution Control Board (CPCB)** formed in 1974; advises the Central Government, coordinates State Boards, sets national standards, collects data and runs training.
- **State Pollution Control Boards (SPCBs)** plan programmes, inspect industries, give consent, lay down effluent standards, monitor and take action.
- Board has a chairman, members from government, local bodies, farming, fishing and industry, and a member secretary.

Main provisions

- **Section 24** no person shall knowingly cause or permit any poisonous, noxious or polluting matter to enter a stream, well, sewer or on land.
- **Section 25** consent (permission) of the State Board is required before starting any industry or treatment system that may discharge effluent - Consent to Establish (CTE) and Consent to Operate (CTO).
- **Section 26** existing industries must also obtain consent; Section 27 :: Board can refuse or give consent with conditions.
- **Sections 20-23** powers to take samples, enter and inspect.
- **Section 33A** power to issue directions, including closure of the industry and stopping of power and water.
- **Penalties** Sections 41 to 44; imprisonment from 1 year 6 months up to 6 years with fine for violations (as amended in 1988). Section 47 :: offences by companies.
- Appeals go to the Appellate Authority / National Green Tribunal.

4. The Air (Prevention and Control of Pollution) Act, 1981

- Enacted to implement the decisions of the Stockholm Conference and to

prevent, control and reduce air pollution. Noise was added as an air pollutant by the 1987 amendment.

- Air pollutant means any solid, liquid or gaseous substance (including noise) present in the atmosphere in such concentration as may be harmful to human beings, living creatures, plants, property or the environment.
- Implemented by the CPCB and State Boards, which have the same structure as under the Water Act.

Main provisions

- **Section 16** functions of the CPCB - to advise the Government, plan nation wide programmes, lay down standards for air quality and provide technical help.
- **Section 17** functions of the State Boards - plan programme, inspect, lay down emission standards for industry and automobiles.
- **Section 19** State Government can declare Air Pollution Control Areas (after consulting the Board) and prohibit use of certain fuels and appliances.
- **Section 21** consent from the State Board is required to establish and operate any industrial plant in a pollution control area.
- **Section 22** no person shall emit air pollutants in excess of the standards.
- **Section 20** control of emissions from automobiles.
- **Section 31A** power to give directions including closure.
- **Section 37** penalty for failure to comply - imprisonment from 1 year 6 months up to 6 years with fine; continuing offence additional fine per day.
- **Section 40** offences by companies.

National Ambient Air Quality Standards (NAAQS 2009) limits for 12 pollutants including SO2, NO2, PM10, PM2.5, ozone, lead, CO, ammonia, benzene, benzo(a)pyrene, arsenic and nickel.

5. The Wild Life (Protection) Act, 1972

- **Objective** to protect wild animals, birds and plants, to control hunting, poaching and trade in wildlife products, and to protect their habitats.
- Gives the legal basis for national parks, sanctuaries and conservation areas. Amended in 1982, 1991, 2002, 2006, 2013 and 2022.
- **Authorities** Indian Board for Wildlife (National Board for Wildlife), Chief Wildlife Warden, wildlife wardens, National Tiger Conservation Authority (since 2006 amendment) and Wildlife Crime Control Bureau.

Main provisions

- Hunting of wild animals in Schedules I–IV is prohibited, except in special cases (danger to human life or disease) with permission.
- **Protected areas** Sanctuaries (limited human activity), National Parks (no grazing or private rights), Conservation Reserves, Community Reserves and Tiger Reserves.
- **Schedules** Schedule I and part II of Schedule II give absolute protection with highest penalty (for example tiger, elephant, lion, rhino, peacock). Schedules III and IV protect less endangered species. Schedule V is for vermin. Schedule VI is for protected plants (for example Kuth, Pitcher plant). (Rationalised in the 2022 amendment.)
- Trade in animal articles, trophies and ivory is banned or licensed.
- Zoos have to be recognised by the Central Zoo Authority.
- **Penalty** for Schedule I offences, imprisonment 3 to 7 years and fine of at least Rs. 10,000 (higher later), more for repeat offences.
- Wildlife areas cannot be changed or de-notified without approval of the legislature and the National Board for Wildlife. The Act is important in EIA because projects near protected areas need Board permission.

Related Act

- **Forest (Conservation) Act, 1980** prior approval of the Central Government is needed to use forest land for non forest purposes; compensatory afforestation is compulsory.

6. Provisions in the EIA Notification (14 September 2006)

- Issued by the Ministry of Environment, Forest and Climate Change (MoEFCC) under the Environment (Protection) Act 1986 and Rule 5(3) of the EP Rules 1986. It replaced the EIA Notification of 27 January 1994.
- Prior Environmental Clearance (EC) is required for new projects and for expansion or modernisation listed in the Schedule (about 39 types) - mining, thermal and nuclear power, river valley, ports, airports, highways, industrial estates, chemicals, cement, pharmaceuticals, townships and others.

Categories of projects

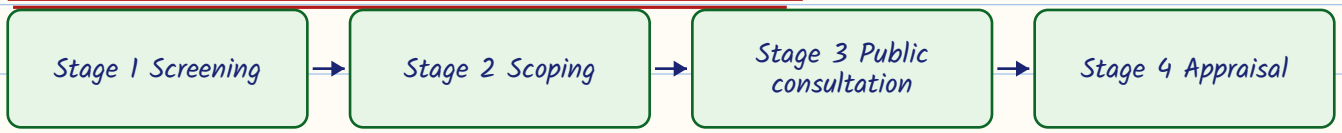
Category	Appraised by	Remarks
Category A	MoEFCC (Delhi) with Expert Appraisal Committee (EAC)	Large and highly polluting projects; EIA and public consultation compulsory
Category B1	State Environment Impact Assessment Authority (SEIAA) with State EAC (SEAC)	EIA report required
Category B2	SEIAA with SEAC	Smaller projects; no EIA report, only Form 1 and conceptual plan

- Category depends on the size, capacity, location (inter-state borders, protected areas, eco-sensitive zones) and type of the project; general conditions apply.
- Expert Appraisal Committees consist of experts in various fields such as environmental science, ecology, engineering and law.
- Some projects do not need public hearing, such as expansion of existing units

without more land and modernisation.

- The validity of the clearance depends on the project (for example 10 years for river valley, 30 years for mining, 5 years for others, later changes).
- Clearance conditions must be followed, and the proponent has to submit half yearly compliance reports.

7. Procedure for Environmental Clearance



Stage 1 – Screening (only for Category B)

- The proponent submits Form 1 (application) with a conceptual plan.
- SEAC decides whether the project is B1 (EIA needed) or B2 (no EIA).

Stage 2 – Scoping

- The EAC / SEAC studies the application and finalises Terms of Reference (ToR) for the EIA report, normally within 60 days.
- Proponent may suggest the ToR; the ToR is displayed on the website. If no reply in 60 days, the proposed ToR is deemed accepted.
- Baseline data collection and EIA study are done on the basis of the ToR.

Stage 3 – Public consultation

- Has two parts: (a) public hearing at the site or near it, and (b) written responses from interested persons.
- The State Pollution Control Board advertises the hearing in two newspapers (one in the local language) at least 30 days before; draft EIA is made available to the public.
- Public hearing is held within 45 days of the request; it is chaired by the District Magistrate or a representative.

- The proceedings are video recorded and sent to the regulatory authority; written comments are accepted for 30 days.
- The concerns of the public are addressed and the EIA report is finalised.

Stage 4 – Appraisal

- The final EIA report with the EMP and public hearing details is submitted to the authority.
- The EAC / SEAC examines it in detail and may ask for more data; appraisal is completed within 60 days.
- The authority takes a final decision (grant, grant with conditions or reject) within 45 days of the recommendation. The total period is 105 days.
- If no decision is communicated in the time, the proponent may presume deemed clearance, as per the rules.
- Grant of EC, with its conditions, is put on the website and in newspapers.



8. Procedure for Conducting the EIA Report

- The EIA study is done by an accredited consultant (as per NABET / QCI scheme).
- Contents as per Appendix III of the 2006 Notification and the ToR :
  - Executive summary and introduction; project description (need, location, size, process, raw materials, utilities, pollution load).
  - Description of the environment (baseline) - air, water, noise, soil, ecology, socio-economic - for one season (other than monsoon).
  - Anticipated environmental impacts and mitigation measures.
  - Analysis of alternatives (technology and site).

  - Environmental monitoring programme.
  - **Additional studies** risk assessment, disaster management plan, social impact assessment and R and R plan.
  - Project benefits.
  - Environmental Management Plan with budget.
  - Summary and conclusion; names of consultants.
- Data is collected by approved laboratories and the report is submitted to the authority and the Board.

9. Evaluation of the EIA Report

- Done by the Expert Appraisal Committee through a review that checks :
  - Whether the report follows the ToR and the proper format.
  - Is the baseline data adequate, recent and reliable?
  - Are all the impacts identified and are the prediction methods suitable?
  - Are the alternatives considered and the site suitable?
  - Are mitigation measures practical and enough? Is the EMP and the budget sufficient?
  - Have public concerns and hearing outcomes been addressed?
  - Does the project comply with laws, standards, and with forest and wildlife clearances?
  - Is the risk and disaster management plan adequate?
- The committee may ask the proponent to submit extra data or revise the report and may visit the site.
- **Final recommendation** approve, approve with conditions or reject.
- **Typical review criteria** completeness, accuracy, objectivity, clarity and usefulness for decision making.
- Public can also raise grievance with the National Green Tribunal (NGT) against

the clearance.

10. Evaluation of Audit Data and Preparation of Audit Report (Recap)

- Audit data (checklists, measurements, records and observations) are organised and compared with legal and company criteria.
- Findings are classed as compliance, non-compliance or observation, and ranked by priority.
- **The report contains** executive summary, introduction, facility description, findings, conclusions, recommendations with action plan and annexures.
- The draft is discussed with the management and then finalised.

11. Post Audit Activities

- Management reviews the report and prepares an action plan - what to do, who will do it, by when, and at what cost.
- **Corrective actions** repair or upgrade of pollution control equipment, training, better record keeping, safe storage of waste.
- **Preventive actions** changes in the process, cleaner production, replacing hazardous materials.
- Follow up audit or verification after a fixed period to check that the actions are completed.
- Reporting to the Board and to the public (Environmental Statement).
- Updating the environmental policy, objectives and the EMS; continual improvement.
- Keeping all documents for future reference.



12. Concept of ISO and ISO 14000

ISO

- ISO = International Organization for Standardization, formed in 1947 with headquarters in Geneva, Switzerland.
- It is a non-governmental federation of national standards bodies from more than 160 countries (India is represented by the Bureau of Indian Standards, BIS).
- **Its standards give uniform specifications for products, services and systems for quality, safety and efficiency. Examples** ISO 9000 (quality management), ISO 45001 (occupational health and safety) and ISO 14000 (environment).

ISO 14000 family

- A series of international standards on environmental management, first published in 1996 by ISO Technical Committee 207. It is a voluntary standard.
- **Purpose** to help organisations minimise harmful effects on the environment, comply with laws and improve continually.

Standard	Subject
ISO 14001	Environmental Management System – requirements (the only certifiable standard)
ISO 14004	Guidelines on EMS principles and support
ISO 14006	Eco-design
ISO 14020 series	Environmental labelling (eco-labels)
ISO 14031	Environmental performance evaluation
ISO 14040 / 14044	Life Cycle Assessment
ISO 14064	Greenhouse gas accounting and verification
ISO 19011	Guidelines for auditing management systems

ISO 14001 (EMS) – key elements

- Follows the Plan – Do – Check – Act cycle.
  - **Plan** environmental policy, identifying aspects and impacts, legal requirements, objectives and targets.
  - **Do** resources, roles, training, communication, documentation, operational control and emergency preparedness.
  - **Check** monitoring and measurement, evaluation of compliance, internal audit, nonconformity and corrective action.
  - **Act** management review and continual improvement.
- Latest version is ISO 14001:2015, which stresses leadership, risk based thinking and the life cycle perspective.

Benefits of ISO 14001

- Better compliance with laws and fewer penalties.
- Saving of raw material, energy and money; less waste.
- Improved image and trust of customers, banks and the public.
- Competitive advantage in the global market.
- Systematic management and continual improvement of environmental performance.
- Certification is given by accredited third party certification bodies after an audit and is valid for 3 years with yearly surveillance audits.

13. Key Points to Remember

- EP Act 1986 is the umbrella Act (Bhopal, Stockholm); Water Act 1974; Air Act 1981; Wild Life Act 1972.
- CPCB (1974) and SPCBs give Consent to Establish and Consent to Operate.
- **EIA Notification 2006** categories A, B1, B2; stages are screening, scoping, public consultation and appraisal.

- **Authorities** MoEFCC with EAC for category A; SEIAA with SEAC for category B.
- Post audit means action plan and follow up audit.
- ISO 14001 is the certifiable EMS standard in the ISO 14000 family.

Important questions (1) Explain the salient features of the Environment (Protection) Act 1986. (2) Water Act 1974 and Air Act 1981 provisions. (3) Wild Life Act 1972. (4) Explain the procedure for environmental clearance as per EIA Notification 2006. (5) How is an EIA report prepared and evaluated? (6) Post audit activities. (7) Write about ISO 14000.